

Academic Qualifications

Year	Degree/Certificate	Institute	CPI
2023-Present	Bachelors of Technology	GLA University, Mathura	8.4
2023	CBSE (XII)	St. Mary's Sr. Sec. School, Dwarka	85%
2021	CBSE (X)	St. Mary's Sr. Sec. School, Dwarka	94.4%

Key Projects

Bol AI Cognitive Intelligence Framework for Voice-Driven System Control | Python, SpeechRecognition, PyAutoGUI (Dec 2025 - Jan 2026)

A modular voice-controlled AI assistant with 40+ commands for productivity, system control, and entertainment

Design & Architecture	- Designed a modular architecture with separated concerns across 7+ modules (voice engine, command handlers, utilities). - Implemented multi-threaded processing for timers and background tasks to ensure responsive user interaction.
Features & Integration	- Built 30+ voice command handlers including Wikipedia search, system control, media playback, and web automation. - Integrated system operations module to control Windows OS (volume, brightness, app launching, process management).
User Experience	- Implemented bidirectional voice interface using pyttsx3 for text-to-speech with customizable voice, rate, and volume. - Created contextual greeting system with time-based responses and personalized assistant naming. - Added productivity features including voice-activated screenshots, calculations, trivia and motivational quotes.

SentinelNexus: Multi-Tenant Enterprise AI Agent | Python, LangChain, LangGraph, OpenAI, Streamlit (Jan 2026 - Jan 2026)

A production-grade AI agent with defense-in-depth security architecture for regulated enterprise environments

Security & Compliance	Architected a 5-layer middleware pipeline implementing PII masking, role-based access control (RBAC), and semantic safety validation
Context Engineering	- Designed dynamic prompt generation system adapting to user clearance levels (1-5 scale), department, and compliance requirements. - Built intelligent memory management with automatic conversation summarization to prevent context degradation.
Access Control	- Deployed Human-in-the-Loop (HITL) middleware requiring manual approval for high-risk operations (server restarts, financial reports) - Created multi-tenant architecture with tenant-level isolation preventing cross-customer data leakage.

Student Performance Prediction | Python, Scikit-learn, CatBoost, XGBoost, Flask, Pandas, NumPy (Jan 2026 - Jan 2026)

An automated machine learning pipeline with ensemble models, hyperparameter optimization, and Flask web deployment for predicting student math scores

Design & Architecture	Implemented custom exception handling with detailed stack trace capture (file name, line number) and structured logging framework for production debugging.
Features & Integration	- Developed 7-model ensemble trainer (CatBoost, XGBoost, Random Forest, Gradient Boosting, Linear Regression, Decision Tree, AdaBoost) with automated model selection. Scikit-learn transformation pipeline with imputation, categorical encoding, and numerical scaling.
User Experience	Flask REST API with HTML form UI for user input and real-time predictions. CustomData class for input validation and seamless feature engineering.

Technical Skills

Core AI & Agentic Frameworks	Machine Learning & Analytics	Backend & Architecture	Infrastructure & Security	Languages & Tools
LangGraph, LangChain, OpenAI API, RAG Pipelines, Context Engineering, Intelligent Memory Management	Ensemble Modeling (CatBoost, XGBoost, Random Forest), Scikit-learn, Feature Engineering, Hyperparameter Optimization	Modular Microservices, REST API (Flask), Structured Logging, Custom Exception Handling	Multi-tenant Architecture, RBAC, PII Masking, Human-in-the-Loop (HITL), Semantic Safety Validation	Python, SQL (Advanced Window Functions), Java, Scala, Streamlit, HTML/CSS, PyAutoGUI

Extra-Curricular Activities

Social	- Volunteered as a teacher for 1 year at an NGO Udaaan Asma Tak funded by GLA University under CSR. - Students from Ajhai Village (nearest village to GLA) are taught by the NGO. Taught students from various classes from 1-12. - Conducted various activities like Children’s Day at GLA Campus for students at Udaaan Asma Tak.
Hackathons	- Won third prize at ML-Manthan Hackathon by GLA University, Mathura for creating a driver alertness detection system. - Participated in the Hack & Viz Hackathon by our college at GLA University. - Participated in ML Hackathon at IIT Kanpur.
Hobby	- Completed Grade-5 from Rockschoool in Electric Guitar - Love to read self-help books - Love to use data to improve my lifestyle